Skills Summary

Algebraic Limits: Factoring and Rationalizing

Use this reference while completing your worksheet or when studying.

Skill 1 — Limits by factoring (limits-by-factoring)

Rule:

If direct substitution gives $\frac{0}{0}$, the numerator and denominator share a factor of $(x-a)$. Factor, cancel it, then substitute.

$a^2-b^2=(a-b)(a+b)$ $\quad$ $a^3-b^3=(a-b)(a^2+ab+b^2)$

Example: $\lim\limits_{x\to 4}\dfrac{x^2-16}{x-4}$

Step 1. Substitution gives $\frac{0}{0}$, so factor rather than conclude.

Step 2. $\dfrac{(x-4)(x+4)}{x-4}=x+4$ for $x\neq 4$; at $x=4$ this is **8**.

Try it: $\lim\limits_{x\to 5}\dfrac{x^2-2x-15}{x-5}$

check: **8**

Skill 2 — Limits by rationalizing

Rule:

When a square root causes the $\frac{0}{0}$, multiply top and bottom by the **conjugate**. $(\sqrt{u}-v)(\sqrt{u}+v)=u-v^2$ clears the radical from the numerator and produces the cancelling factor.

Example: $\lim\limits_{x\to 0}\dfrac{\sqrt{x+1}-1}{x}$

Step 1. Multiply by $\dfrac{\sqrt{x+1}+1}{\sqrt{x+1}+1}$ so the numerator becomes $(x+1)-1=x$.

Step 2. Cancel x to get $\dfrac{1}{\sqrt{x+1}+1}$; at $x=0$ this is **1/2**.

Try it: $\lim\limits_{x\to 0}\dfrac{\sqrt{x+9}-3}{x}$

check: **1/6**

Skill 3 — Removable discontinuities

Rule:

A piecewise f is continuous at $x=a$ exactly when $\lim_{x\to a}f(x)=f(a)$. To patch a hole, set the assigned value equal to the limit you computed.

Example: $f(x) = \dfrac{x^2 - 100}{x - 10}$ for $x \neq 10$, $f(10) = c$. Find c.

Step 1. Cancel: $\dfrac{(x-10)(x+10)}{x-10} = x + 10$, so the limit at 10 is 20.

Step 2. Continuity forces $f(10)$ to equal that limit, so $c = \mathbf{20}$.

Try it: $f(x) = \dfrac{x^2 - 25}{x - 5}$ for $x \neq 5$, $f(5) = c$.

check: **10**

(!) Watch out: Cancelling changes the function at the removed point, not the limit. Never write $\frac{0}{0} = 1$, and never conclude a limit fails to exist just because substitution is undefined.